

IBONG TIRIRIT (PIPE 8)

- A type of condenser combines the functions of condenser and cooling tower is called

Evaporative condenser

- An expansion valve with fluid charge remote valve that maintains a constant degree of superheat in the evaporator is called

Thermostatic expansion valve

- A refrigeration system ideally suited for use in air craft because it is light in weight and requires less space is called

Air Cycle

- Refrigerant with chemical formula CHClF_2 is designated by

R-22

- Refrigerant with chemical formula $\text{CF}_3\text{CH}_2\text{F}$ is designated by

R-134A

- Carbon Dioxide refrigerant is designated by

R-744

- Piping, fittings and valve materials found suitable for use in Ammonia refrigerant are manufactured from

Iron and Steel

- Which statement is incorrect with Water-Cooled Shell-and-Tube type of Ammonia

Water flowing through passes inside tubes and the refrigerant condensing in the shell

- In Vapor Absorption Refrigerant system utilizing ammonia as refrigerant and the absorbent is

Water

- The most common secondary refrigerant medium used in indirect refrigeration system such as in industrial ice plants and cold storages

Brine

- A constant restriction expansion device in refrigeration system which is a long tube with narrow bore is called

Capillary Tube

- Mixtures of refrigerants that which behave like pure substance are called

Azeotropes

- In refrigeration compression that has an electric motor and the compressor built into an integral housing is commonly called

Hermetic Compressor

- Compounds with the same chemical formula but different molecular structure such R-134A and R-134 are called

Isomers

- An expansion valve used for flooded type of evaporator in a refrigeration system is called

Float Valve

- An atom in the refrigerant molecule consider responsible for the depletion of ozone layer in upper atmosphere (stratosphere)

CL Atom

- A vapor compression system used for production of low temperature in which a series of refrigerant with compressively low boiling point are used in series of single stage units is called

Cascade System

- Which statement is correct on the effect of liquid subcooling in a vapor compression system

Subcooling decreases the flashing of the liquid during expansion and increase the refrigerating effect

- Which statement is correct on the effect of suction vapor superheating in a vapor compression system

Superheating both increases specific compressor work and refrigerating effect

- Which statement is correct on the effect of increasing condensing pressure in a vapor compression system.

Increasing condensing pressure increases specific compressor work and decreases the refrigerating effect

- Freezing method utilizes the combined effect of low temperature and high air velocity to produce a high rate of heat transfer from products

Blast Freezing

- A component installed a refrigeration system to ensure that no liquid enters the compressor and subcooled the liquid from condenser to prevent bubbles of vapor from impeding a flow of refrigerant through expansion valve is called

Liquid to Suction Heat Exchanger

- Refrigerant used in steam jet refrigeration is

Water

- Which refrigerant is completely miscible with oil

R-12

- Ammonia is not used in domestic refrigeration and comfort air conditioning because

Ammonia is flammable and toxic

- Refrigerant-22 which is an HCFC has $1/20^{\text{th}}$ the ozone depletion potential (ODP) of R-11 and R-12, hence it can be continue to be use for quite sometime however all these have global warming potential (GWP) because of this R-22 will have to be phase out by year

2030 AD

- Which of the following is incorrect regarding the function of the liquid receiver

Stores oil that is carried along the refrigerant

- Shut-off valve that is actuated by an electro-magnetic coil is called

Solenoid Valve

- A very important consequence of ideal gas model is that the internal energy of an ideal gas is a function of ____ only

Temperature

- At what temperature do the Fahrenheit and Celsius scales have the same value

-40°

- States that if the temperature of a given quantity of gas is held constant, the volume of the given gas varies inversely with absolute pressure during the change of state

Boyle's Law

- A unit of force that produces unit acceleration in ft/sec^2 and in a unit of mass in lbm.

Poundal

- Why does a cube of ice float in water?

The density of ice is lesser than water

- The work done in fusion a fluid across a boundary usually into or out of the system is called

Flow Energy

- States that energy is neither created nor destroyed but it can only change in forms

1st Law of Thermodynamics

- Heat is defined as form of energy that is transferred between two system by the virtue of ____

Temperature Difference

- The usual energy transferred in BTU/hr provided by input of power in Watts, this is just the coefficient of performance expressed in mixed units

Energy Efficiency Ratio (EER)

- If EER is the energy coefficient ratio the COP is the coefficient of performance then

EER = 3.41 COP

- Which of the following is the reverse rankine vapor cycle

Vapor Compression Cycle

- The most common type of refrigeration application in household refrigerators, air conditioning for cars and houses, chillers and so on

Vapor Compression Cycle

- Which of the following is a reversed Brayton Cycle

Air Refrigeration Cycle

- Which of the following cycle is bulky and involves toxic fluids hence it is unsuitable for home and autopulling

Absorption Cycle

- Which of the following refrigeration cycle is practically when large quantity of waste or expensive heat energy area available

Heat-Driven Refrigerating Cycle

- A product of computer aided manufacturing and precision machining where introduced commercially in the late 1980's as replacement for reciprocating compressors in small residential air conditionings

Scroll Compressor

- Condenser used in small and medium size up to approximately 100 tons of refrigerators

Air-Cooled Condensers

- For efficient operation the condensing temperature should not be lower than

5° C

- For efficient operation the condensing temperature should not be more than

17° C

- Condenser used for larger capacity refrigerators

Water-Cooled Condensers

- Coolers and chillers for water generally operates with an average temperature difference of

3°C to 11°C

- To avoid freezing problems, entering refrigerant should be

Above -2°C

- Suction line should not be filled to large as reasonable velocity which needed to carry oil from the evaporator back to the compressor for horizontal suction lines the recommended minimum velocity is

3.8 m/s²

- Type of turbine used for low heads, high rotational speed and larger flow rate

Axial Flow Turbine

- To keep the deflected jet out of the way of the incoming jet, the actual angle is limited to approximately

165 degrees

- Which of the following turbines are centrifugal pumps operating on reverse

Reaction Turbine

- Also known as Francis Turbine or Radial Flow Turbine

Reaction Turbine

- The turbine draft tube and all related part comprise what is known as

Setting

- When forebay is not part of generating plant design, it will be desirable to provide a _____ in order to relieve the effect of rapid changes in flow rate

Surge Chamber

- To keep the deflected jet out of the way of the incoming jet, the actual angle is limited to approximately

165%

- The minimum fluid energy required at the pump inlet for satisfactory operation is known as

NPSHR

- Throttling the input line to a pump and a venting evacuating the receiving pump

Both increase cavitation

- Traditional reciprocating pumps with pistons and rod can be either single acting or double acting and or suitable up to approximately

2,000 psi

- Plunger pumps are only single acting are suitable up to approximately

10,000 psi

- The ratio of the actual to the ideal heat transfer coefficient

Cleanliness Factor

- The ratio of the maximum and minimum mass stream flow rates at with the temperature can be accurately contained by the desuperheater

Turndown Ratio

- With the reversible generator, the thermal efficiency of ericsson cycle is _____ to that of the carnot cycle

Equal

- Use to describe the act of flowing the exhaust product out with air fuel mixture

Scavenging

- Which of the following compresses and increases the amount of air that enters the cylinder per stroke

Supercharging

- A form of supercharging in which the exhaust gases drive the supercharger

Turbocharging

- A close heat exchanger that transfer heat from compressed air to cooler air

Aftercooler

- At the control stand, each of the turbine blades must be adjusted manually by a

Sensor

- The most commonly used adjusted blade propeller turbine

Kaplan Turbine

- Runner blades are adjusted automatically in synchronization with turbine wicket gates by

Oil Servomotor

- What is applied to the servomotor from the turbine governor oil system through a generator shaft and through a control valve

Oil

- Used to exhaust air from the casing for starting

Ejectors

- Advantageous for propeller turbine of all types

Siphon Setting

- Turbine setting where one disadvantage is a difficulty of lubricating operating mechanism with consequent relatively rapid wear

Open Flume Setting

- A patent device combining expansion and compression function in a single rotor permitting higher cycle temperature

Comprex

- A rough measure of physical sides of the equipments with must handle the specified quantity of the fluid

Air Rate

- The ratio of the highest mean compressor discharge pressure to the lowest mean compressor inlet pressure

Cycle Pressure Ratio

- The ratio of the maximum pressure in the cycle to the atmospheric pressure

Cycle Pressure Level

- The ratio of the actual partial pressure exerted by the water vapor in any volume of air to the partial pressure that would be exerted by the water vapor if the water vapor in the air is saturated at the temperature of the air

Relative Humidity

- An expression of the mass of water vapor per unit mass of dry air is

Humidity Ratio

- Which of the following is equivalent to 1 lb

7,000 grains

- The locus that states that have the same value of stagnation enthalpy and mass flux is called

Fanno Line

- Combining the conservation of mass and momentum equation into a single equation and plotting it on the h-s diagram yield a curve called

Rayleigh Line

- Across the shaft the stagnation temperature of an ideal gas

Remains Constant

- The locus of point where condensation will take place regardless of initial temperature and pressure at the nozzle entrance

Wilson Line

- The enthalpy of a substance at specified state due to its chemical composition

Enthalpy of Formation

- Since the air-conditioning apparatus usually removes heat and moisture from both the conditioned room and from outside make-up air the sensible heat ratio that can be used is

Grand Sensible Heat Ratio

- The slope of the line between the apparatus dew point on the saturation line and to design conditions of the condition space

Effective Sensible Heat Ratio

- The sensible heat ratio is a

Psychrometric Slope

- The percentage of the air that is not cold or heated is

Bypass Factor

- The bypass factor for large commercial units

Around 10%

- The bypass factor for residential units are approximately

35%

- The compliment of the bypass factor is

Coil Efficiency

- The mass of water sprayed to the mass of air passing through the washer per unit time

Spray Ratio

- In a cooling tower, the water passes over staggered flats or interior fills also known as

Packing

- The ratio of total dissolve solids in the re-circulating water to the total dissolve solids in the make-up water

Ratio of Concentration

- Through windage removed some of the solids most must be removed by bleeding some of the water off, this is known as

Drift

- A Tri-cooling tower where steam travels through large diameter 'Trunks' to a cross flow heat exchanger where it is condense at cooling by the cooler air

Direct Condensing Tower

- Which of the refrigerant of choice in uniting air-conditioners?

R-22

- In new equipment which of the following replaces R-11?

R-123

- New combustion turbines on cutting edge of technology (advance turbine system) are available to achieve

38 to 38.5%

- How many percent of solar energy survives absorption and reflection?

40-70%

- Which of the following collect used mirrors and over or lenses to focus the suns energy in a small absorber area?

Concentrating collectors

- Which of the following collectors are useful when extremely hot transfer fluid is needed and or generally limited to commercial project

Evacuated tube collectors

- The ratio of energy absorbed by the transfer fluid to the original incident energy striking the collector

Collector efficiency

- The shading factor in calculating the heat absorbed by the scalar collector has a value approximately

0.95 to 0.97 %

- Which of the following approaches to the solar energy generating systems describe the parabolic tracking through concentrators focus sunlight on evacuated glass tubes that run along the collectors focal line

Through electric system

- Through electric energy technology is relatively mature but due to the low temperature, average annual thermal efficiency are only

10-15%

- A filled of heliostats or tracking mirrors concentrates solar energy onto a receiver on a central tower

Power tower system

- In a power tower system typical thermal efficiency is in the range

15 to 20%

- A flash steam cycle can be used in hot water temperature is approximately

165 degrees C or higher

- A binary cycle using separate heat transfer fluid which the temperature of the cooling water is between approximately

165 and 120 degrees C

- In hot rock system water is injected through injection wells into artificially made fractured rock of how many km below the surface

1 to 6 km

- What temperature is required to ignite the fuel oil?

800 to 1000 degrees F

- What air pressure is required to produce the required ignition temperature

350 to 500 psi

- The portion of the piston which extends below the piston pin and serve as a guide for the piston and connecting rod

Piston skirt

- Binding of the piston and the cylinder wall as the result of the lubrication having been destroyed by excessive temperature and friction

Piston seizure

- Combination of liquids which do not mixed or combined chemically

Emulsion

- At what temperature will self igniting cartridge ignite

About 190 degrees F

- A fuel is supplied by a single pump and switch to which cylinder by a multi outlet rotating valve or distributor

Distributor system

- A chamber so designed that injection takes place directly opposite its outlet the chamber extending backward from the outlet

Ante combustion chamber

- A cylindrical vessel connected in the fuel system to absorb the shaft of the injection pumps and to provide a reservoir at high press for supply an even flow of fuel oil to the cylinders

Accumulator

- Which of the following is used to improve cold weather starting?

Glow Plug

- The rated power that the manufacturer claims to engine is available to provide or a continuous basis without incurring damage

Continuous Duty Rating

- The peak power that can be produced on an occasional basis

Intermittent Rating

- A value of a property that includes the effect of friction is known as:

Brake Value

- Forced draft fan are run at relatively high speeds the range of.

1,200 to 1,800 rpm

- is that rely on natural draft are sometimes referred to as:

Gravity Chimney

- For realistic problems the achievable stock effect probably should be considered to be.

80% of the ideal

- In a balance system, the available draft is.

0 (zero)

- It is a device used for atomizing or cracking fuel oil and through which a fuel oil is injected into the working cylinder of diesel engine.

Atomizer

- What is the proper seat width of a spray valve

1/16 in.

- A device which automatically governs or controls the speed of engine.

Governor

- The term enthalpy

Will measure by its change, the quantity of heat added when refrigerant is vaporized at constant pressure.

- When heat is added to moisture air in an air-conditioning process.

The absolute humidity remains constant.

- Air having a relative humidity of 50% is heated in an air conditioning apparatus after having it will be found that

Its absolute humidity remains constant.

- The ton of refrigeration is a heat unit equivalent to

12,000 Btu/hr

- In the process of heating atmospheric air in an air conditioning apparatus

The absolute or the specific humidity

remains constant.

- The temperature at which the vapor in the mixture is cooled at constant pressure is called

Dew Point Temperature

- Refers to the amount of air that has not been in contact during the sensible cooling process in the surface of the cooler coil

Coil Bypass factor

- The process that the body uses to convert energy into heat and work or the process that determines the rate at which energy is converted from chemical to thermal form within the body

Metabolism

- It is a body in insulation that is usually describe as a single equivalent uniform layer over the whole body.

Clothing

- It is the basic index used to describe the radiant conditions in a space. It is the mean temperature of individual exposed surfaces in the environment

Mean Radiant Temp

- It is the most common instrument used to determine the mean radiant temp it consist of a hollow sphere 6 inches in diameter flat black paint coating and a thermo-couple or thermometer valve at its center

Vernon's Globe Thermometer

- It is the mechanism of heat transfer between parts of a continuous because of transfer of energy between particles or group of particles the atomic level, it is the mode of heat transmission through the building structure like the wall due to the temperature diff. between surface of the wall

Thermal conduction

- It is the transport of energy by mixing in addition to conduction. It is associated with fluids in motion generally through a pipe or duct along a surface

Thermal Convection

- A form of thermal convection when the bulk of the fluid is moving relative to the heat transfer surface because such motion is usually caused by a blower fan or pump that is forcing the flow

Forced Convection

- A form of thermal convection when the motion of the fluid is due entirely to buoyancy forces usually confined to a layer near a heated or cooled surface.

Free convection

- A mode of heat transfer by the electro magnetic waves an entirely diff. phenomenon from conduction and convection it can occur in perfect vacuum and it's usually impeded by an intervening medium

Thermal Conduction Radiation

- An envelope area with significantly higher rate of heat transfer than the contiguous enclosure is called

Air Conditioned room

- It is the total thermal radiation that impinges on a surface from all directions and from all sources

Total or Global irradiation

- Transformation of the radiant energy into thermal energy stored by molecules is known as

Radiation

- It is the return of radiation by a surface w/c change of surface frequency the radiation is bounced off the surfaces

Reflection

- In a refrigeration system the heat absorb in the evaporator per kg of refrigerant passing through

Equals the increase in enthalpy

- It is the passage of radiation through a medium w/o change of frequency

Transmission

- It refers to any glaze aperture in a building envelope the components of each include glazing material either glass or plastic flaming mull ones, muntins and dividers external shading devices and integral shading system between glass

Fenestation

- The rate at which energy is transferred to or generated within a space is called

Heat Gain

- It is the rate at which energy must be removed from a space to maintain the temp. and humidity at the design values

Cooling Load

- It is the rate at which energy is removed from a space by cooling and dehumidifying equipment. This rate must be equal to the cooling load when the space conditions are constant when the equipment is operating

Heat extraction rate

- It is a computer or hand circulation procedure where energy requirement are determine at many outdoor temp. conditions. It is used for the energy calculations of air conditioning system

Bin Method

- It is the temperature of uniform environment at 100% relative humidity in which a person or occupant losses the same total amount of heat from the skin as in the actual environment

Humid Operative Temp.

- It is the ratio of the total evaporative heat loss required for the thermal equilibrium to the maximum evaporative heat loss possible to the environment multiplied by 100 for the steady-state conditions and with the skin temperature constant at 95%

Heat Stress Index

- It is the temperature of a uniform environment at 0% relative humidity in which a person or occupant causes the same total amount of heat from the skin as in the actual environment

Adiabatic Equivalent Temperature

- Heat added to or removed from a substance that cause a change in temperature

Sensible Heat

- It is an environment heat stress index that combines the dry bulb temperature a naturally ventilated wet bulb temperature and the globe temperature

Wet Bulb Globe Temperature

- It is an empirical index for the combined effect of wind and low temperature

Wind Chill index

- A method of air handling to air conditioned room that produces more economical performances as compared with system where heat might be required is called

Recirculated Air with External Bypass

- It is the ratio of the air humidity ratio to the humidity ratio of saturated air at the same temperature

Percent Saturation

- It is the ratio of the actual partial pressure of vapor and the saturation pressure of vapor corresponding to the actual temperature of the mixture

Humidity Ratio

- Heat gain or Heat loss due to the temperature diff. across the building element

Transmission

- 1 ton of refrigeration is equivalent to
12000 btu/hr
50.4 kcal/min
200 btu/min
All of the Above

- Air whose condition is such that any decrease in temperature will result in condensation of the of the water vapor into liquid

Saturated Air

- The binary mixture of dry air and water vapor is known as

Moist Air

- In compressing air in water jacketed air compressor the power required to drive it

depends on the temperature change of the air for a given pressure range.

- An instrument used in the study of the properties of air

Hydrometer

- The difference between dry bulb and wet-bulb temperature

Wet bulb depression

- The transmission of heat from one place to another by fluid circulation between spots of different temperature is called

Convection

- A machine or equipment used for drying process

Dryer

- Water occurring when chemical component of the material changes its chemical composition by heat or other means is called

Chemically Combined Water

- A Wet type mechanical classifier (solid separator) in which solids settle out of the carrier liquid and are concentrated is known as

Dewaterer

- A term used for describing the moisture in the mass that is not on the surface of material

Inherent Moisture

- The main principle of the term drying process

Evaporation

- The common term for process using only pressure suction or decantation

Dewatering

- The process of removing moisture in varying amount from solids or semi fluid material. The process may be accomplished by pressure suction decantation or evaporation

Drying

- A type of dryer that may have to flame from combustion impinging on the material being dried or the gases of combustion maybe mixed with additional air so that mixture in contact with the material is reduce in temperature

Direct Type Dryer

- The most commonly used dryer that consists of a rotating cylinder inside each the material so while so while

getting in contact with hot gasses, the cylinder is tilted at the right angle and fitted with lifting flights. This drier is used for copra, sand or wood chips

Rotary Drier

- A dryer where the material is in contact with steam pipes or the air is passed over steam heaters and then over or to the material being dried

Steam Heated Dryer

- A dryer that consists of a centrifuge revolving at a high speed causing the separation by centrifugal force of the water from the material. This dryer is used for drying fertilizer, salt and sugar

Centrifugal Dryer

- A Dryer type in which material to be dried is supported on a floor through which the hot gasses passed, this drier is used for copra, coal and enamel wares

Hearth Dryer

- Those substance that are particularly variable in the moisture content that they can passes at different times

Hygroscopic Materials

- The hygroscopic moisture content of a substance expressed as a percentage of the bone dry weight of the material

Regain

- A dryer that consists of a trace carrying the material to be dried placed in compartment or moving conveyor, this type of dryer is used for ipil-ipil leaves and grains

Tray Dryer

- A type of dryer that consists of a vertical shaft in which the wet feed is introduced at the top and falls downward over baffles while coming in contact with the hot air that rises and exhaust at the top. This type of dryer is used for drying palay wheat and grains

Tower Dryer

- The temperature to which the air must be cooled at constant pressure to produce saturation is called

Dew Point

- The mass of water vapor per unit volume of air

Absolute Humidity

- When the expansion or compression of gas takes place without transfer of heat to or from the gas the process is called

Adiabatic

- Weight per unit volume is termed as

Weight Density

- The designed of an air supplied duct of an air conditioning system

Affects the distribution of air

- Atmospheric pressure at sea level is

14.7 psi

- Air contains by weight

23% O₂

- An instrument used to determine the specific gravity of a solution

Hydrometer

- The type of heat that changes the physical state of a substance

Latent Heat

- Refers to the temperature at which all molecular motion ceases according to the kinetic theory of heat

Absolute Zero

- 1 ton of refrigeration is a heat equivalent to

12 000 BTU/hr

- Is one whose temperature is below the saturated temperature of 1 gram of water to 1 degree?

Sub cooled

- Weight per unit volume is termed as

Weight Density

- Measure of hotness or coldness of the body

Temperature

- The temperature at which the water vapor in the air begins to condense or the temperature of which the relative humidity of air becomes 100%

Dew Point

- It consist of a wet and dry bulb thermometers mounted on a strip of a metal

Sling Psychrometer

- A manometer is an instrument that is used to measure

Air Pressure

- The relationship of water vapor in the air at the Dew Point temperature to the amount in the air if the air is saturated at the dry bulb temperature is

Partial Pressure Actual Dew Point

- All of the following statements about WB temperature are true except

WB temperature lies numerically between dew point and dry bulb temperature for unsaturated system

- Heat added to or remove from a substance that cause a change of temperature

Sensible Heat

- In a sensible heating process, the final humidity ratio is

The same

- In an adiabatic saturation process, the final relative humidity is

Increase

- Cooling and dehumidifying process is commonly used for

Summer Air-Conditioning

- An adiabatic saturation process is also known as

Constant WB temperature process

- The process of simultaneous heating and dehumidifying is known as

Chemical Dehumidifying

- The process of increasing the dry bulb temperature without changing the humidity ratio is known as

Sensible Heating

- The process of cooling without changing the humidity ratio is known as

Sensible Cooling

- The process of simultaneous heating and humidifying is known as

Heating and Humidifying

- The process of cooling and maintaining the temperature of space or material below its surrounding

Refrigeration

- The process of treating air so as to control simultaneously its temperature humidity cleanliness and distribution to meet the requirement of the conditions phase

Air Conditioning

- An appliance used to dehumidify air in homes and buildings which uses refrigerant unit

Dehumidifier

- A component of vapor compression refrigerant system which is used to compressed refrigerant vapor from evaporator to the condenser

Compressor

- One of the components at the high pressure side of the vapor compression refrigerant system

Condenser

- The thermal state of a body considered with reference to its ability to communicate heat to the other bodies

Temperature

- A form of energy that flow at higher temperature body to lower temperature body

Heat Body

- A thermodynamic quantity that measured the fraction of the total energy of a system that is not available for doing work

Entropy

- The total volume of a mixture of gases is equal to the sum of the volume that would be occupied by the various component each at the pressure and temperature of the mixture

Amagat's Law

- The total pressure of a mixture of gases is the sum of the pressure that each gas would exert for it to occupy the vessel alone at the volume and temperature of the mixture

Dalton's Law of Partial Pressure

- The ratio of the actual partial pressure of vapor and saturation pressure of vapor corresponding to actual temperature of the mixture

Humidity Ratio

- Steady flow process at total pressure to a control volume for which there is no heat and in which a liquid and the gas flowing to a right at a local equilibrium state containing saturated vapor before the exit section is reached

Adiabatic Saturation Process

- The actual temperature of air which is measured by an ordinary thermometer or the temperature of the gas or mixture of gases indicated by an accurate thermometer after correction for radiation

Dry Bulb Temperature

- An instrument of two thermometers one used to measure the dry bulb temperature and other one used to measure wet bulb temperature of air

Barometer

- The ratio of the mass of water vapor and the mass of dry air in the mixture

Humidity ratio
specific humidity
mixing ratio
All of the Above

- The sum of the internal energy and the product of pressure and specific volume

Enthalpy

- The temperature at which the condensation of water vapor in space begins

Dew Point Temperature

- The temperature of moist air which is equal to the saturation temperature of the partial pressure of the vapor

Dew Point Temperature

- The ratio of the air humidity ratio to the humidity ratio of saturated air at the same temperature

Percent Saturation

- At constant temperature the absolute pressure of gas varies inversely as its volume

Boyle's Law

- At constant pressure the volume of gas is proportional to its absolute temperature at constant volume the pressure is proportional to its absolute temperature

Charle's Law

- A part of refrigerating unit in which refrigerant vaporizes to absorb heat

Evaporator

- A part of refrigerating unit in which refrigerant undergoes transformation from gas to liquid

Condenser

- If the dry and wet bulb temperature readings of the sling psychrometer are identical

The air is saturated

- In an adiabatic saturation process the relative humidity is

Increase

- The temperature at which the water vapor in the air begins to condense or the temperature at which the relative humidity of the air becomes 100%

Dew Point Temperature

- Refers to the temperature at which a molecular motion ceases according to the kinetic theory of heat

Zero Absolute Temperature

- Heat gain or heat loss due to the temperature difference across a building elements

Transmission

- Heat gain due to transmission of solar energy through a transparent building component or absorption by an opaque

Solar

- A mixture of oxygen and nitrogen plus small amount of rare gases such as argon is known as

Air

- What is the insulating value of clothing

Clo Units

- If two identical centrifugal pump are installed in series, the system flow rate

Remains the same

- The formation at collapse of vapor bubbles that occurs when pump inlet suction pressure falls to or below the vapor pressure of the liquid is called

Cavitation

- The process done in centrifugal pump by a filling up the suction line and pump casing with liquid to removed air or vapors from the water waste of the pump is called

Priming

- Which term does not belong to the group

Diaphragm pump

- This is a type of reaction turbine usually with typical spiral casings where water enters the runner radially at its outer peripheral and changes directions (imparting energy to the runner) while flowing along the complex shape profiles to discharge axially it is the most widely used type of hydraulic turbine for modern powered generation process

Francis Turbine

- It is the expanding tube connecting the outlet passage of turbine with the tail water

Draft tube

- It is a hydro-electric plant in which excess water pump to an elevated space during off peak period and the stored water will be used to dry hydraulic turbine during the peak period to meet peak demand

Pump Storage Plant

- It is a channel that conducts water away from the turbine

Tail Race

- This is a grid or screen compose of parallel bars to catch floating debris it prevents leaves broaches and other water contaminant from entering the penstock

Trash Rack

- It is the term used that refers to water in the reservoir

Head water

- This is a propeller type reaction turbine with both adjustable guide vanes and runner blades

Kaplan Turbine

- It is basically hydro-electric power utilizing the difference in elevation between high and low tide to produce energy

Tidal Power

- The power output of the plant depends on the water flow in the river at low river flows, the output decrease accordingly

Run-of-River Plant

- The intake pipe to a hydraulic turbine to a dam

Penstock

- A kind of fluid flow where in the fluid travels parallel to the adjacent layers and the paths of individual particles do not cross or intersect

laminar Flow

- Conducts the water around the turbine

Spiral Casing

- These are removable vertical vanes that are actuated by the governor to control the flow of water and therefore the energy supplied to the runner

Wicket gate

- The speed of hypothetical model turbine having the same configuration as the actual turbine when the model would be of the proper size to develop 1 HP at the head of 1 ft

Specific Speed

- The difference in elevation between the head water and tail water or tail race

Gross Head

- The ratio of the effective head to the gross head

Penstock Efficiency

- The ratio of the head utilized to the effective head

Hydraulic Efficiency

- Heads between 70 and 800 ft generally indicate this type of turbine

Reaction Turbine

- Heads below about 70ft fall this type of turbine

Propeller Turbine

- The Functions of this hydraulic turbine part are (1) it enables the turbine to be set above the tail water level without losing any head thereby and (2) it reduces the head loss at the merge discharge to increase the net head of available to the turbine runners

Draft Tubes

- It is a tank with free surface provided at the transition from the low pressure head raise or tunnel to the penstock it protects the head raise pipe or tunnel from excessive changes in pressure in supplying or storing water as required

Surge tank

- The rotating part of the turbine where the water impart its energy onto the turbine shaft

Runner

- The information and collapse of vapor bubbles that occurs when the pump inlet suction pressure falls to or below the vapor pressure of the liquid is called

Cavitation

- A property of lubricating oil that measures the thickness of the oil and will help determine how long oil will flow at a given temperature is known as

Viscosity

- A flow at low Reynolds number with smooth stream lines and shear and conduction effects flowing entirely to the fluids molecular viscosity and conductivity

Laminar

- The fact that the buoyant force of the floating object is equal to the weight of displaced liquid is known as

Archimede's Principle

- The primary purpose of a turbine in a fluid loop is to

Extract Energy from the flow

- It provides an efficient safe means on releasing flow water that exceeds the design capacity of the dam

Spill way

- The power required to deliver a given quantity of fluid against a given head with no losses in the pumps

Hydraulic Power

- The velocity of fluid is zero at the wall and the maximum at the center because of the

Viscous effect

- The work termed for pumps, compressor, fans and blowers is negative since work

Done on the fluid

- This is a type of water turbine where a jet of water is made to fall on the blades or buckets and due to the impulse of water, the turbine will start moving

Pelton wheel turbine

- A device used to prevent water hammer in turbine

Surge tank

- It is a vent or hole in the earth surface usually in volcanic region from which stream gaseously force or hot gases issue

Fumarole

- The typical depth of geothermal production well in meters

3000

- A wind energy sys transforms the _____ of the wind into mechanical or electrical energy that can be harness for practical used.

Kinetic Energy

- It is the rhythmic rise and fall of the ocean water

Tide

- The power available in the wind is proportional to the _____ of its speed

Cube

- It is the maximum tidal range

Spring Tide

- Natural gas is considered _____ when it is almost pure methane

Dry

- In this type of reactor the water is heated by the nuclear fuel and boils to steam directly into the reactor vessel it

is then pipe directly to the turbine, the turbine spins driving the electrical generator producing electricity

Boiling Water Reactor

- It is an air pollution control device that works by electrically charging the particles of fly ash in the flue gas and collecting them by attraction to charge metal plate

Electro static precipitator

- This nuclear reactor component which is made up of carbon and beryllium slows down the fast neutrons that are born during the fission process

Moderator

- This boron coated steel rods are used to regulate the rate of fission chain reaction they are withdrawn from the core to start a chain reaction and inserted all the way into the core to stop it

Control rods

- In general usage the term combine cycle plant describes the combination of a gas turbines generator _____ cycle with turbine exhaust waste boiler and steam turbine generator (Rankine Cycle) for the production of electric power

Brayton Cycle

- Natural gas is a fossil fuel formed when the layers of buried plants and animals are exposed to intense heat and pressure. It has been club as the fuel of the future or green fuel and comprises mainly of

Methane

- If fall ash that exits the combustion chamber in the flue gas and is captured by air pollution control equipment such as hydrostatic precipitators, bag houses and wet scrubbers

Fly Ash

- Also known as brown coal, it is the lowest rank of solid coal with the calorific value of less 8300 Btu/lb on a moist mineral matter free basis

Lignite

- It means using the same energy source for more than one purpose such as using the waste heat from an engine for space heating

Co - generation

- The minimum amount of air required for the complete combustion of fuel

Stoichiometric Air

- The device or instrument used for measuring the calorific value of a unit mass of fuel is called

Calorimeter

- When water in the products of combustion is in the vapor or gaseous form the heating value is known as

Lower Heating Value

- The percent excess air is the difference between the air required divided by

The Theoretical air supplied

- During the combustion process the components which exists before the reaction are called

Reactants

- The minimum amount of air needed for the complete combustion of fuel oil is

Theoretical Air

- A gas which will not be found in the flue gasses produced from the complete combustion of fuel oil is

Hydrogen

- The higher heating value is determined when the water in the product of combustion is in

Liquid form

- The amount of heat liberated by the complete combustion of a unit weight or volume of fuel is known as

Heating Value

- An atemporator is another name for

Desuperheater

- What is the fundamental indicator of good combustion

Presence of minimum oxygen, maximum carbon dioxide and nitrogen combustibles in the flue gas

- It is a solidified mass of fused ash

Carbon Residue

- Why does older type of economizers where constructed in variable of cast iron

Because cast iron resist corrosion better than mild steel and the pressure where comparatively low

- It is an accessories often installed on modern boilers to preheat air for combustion before it enters the boiler furnace

Air Preheater

- What component are included in the approximate analysis in solids fuel

Fixed Carbon, Volatile Matter, Ash and Moisture

- Which element of fuel is not combustible

Oxygen

- A device or an instrument used to recover the cylinder pressure of an engine piston travel in an x-y graph where pressure forms the vertical axis and piston travels forms the horizontal axis

Engine Indicator

- This diagram is used by the operating engineer to determine leaky piston packing, sticking piston, incorrect valve timing, loosing bearings restricted and or outlet piping, etc.

Indicator Diagram

- The effective weight of the brake arm when the brake bond is lose

Tare Weight

- The power output of the generator

Electric Power

- The amount of fuel needed to perform a unit of power

Specific Fuel Consumption

- The ratio of heat converted to useful power to the heat supplied

Thermal Efficiency

- The ratio of the actual power of the engine to its ideal power

Engine Efficiency

- The ratio of the cylinder volumes after and before the combustion process

Cut - off Ratio

- The position of the piston when it forms the smallest volume in the cylinder

Head End Dead Center Position

- The position of the piston when it forms the largest volume in the cylinder

Crank End Dead Center Position

- When four events takes place in one revolution of a crank shaft of an engine, the engine is called

Two stroke cycle engine

- What is the model cycle for spark ignition engine?

Otto Cycle

- Diesel Engine fuel is rated in terms of:

Cetane No.

- The distance that the piston can travel in one direction

Stroke

- The minimum volume formed in the cylinder when the piston is at the top dead center.

Clearance Volume

- It is a fictitious pressure if it acted stroke would produce the same amount of network as that produce during the actual cycle.

Mean Effective Pressure

- The ratio of constant pressure specific heat to the constant volume specific heat.

Specific Heat Ratio

- Ideal Cycle of the compression ignition reciprocating engines.

Diesel Cycle

- The power develop in the engine cylinder as obtained from the pressure in the cylinder.

Indicated Power

- The pressure and torque spent in over coming friction of reciprocating and revolving parts of the engine and the automobile before it reach the drive shaft.

Friction Power

- It is an instrument for determining brake power usually by independent measurement

of force, time and distance through which the force is moved.

Dynamometer

- It is the indicator used to determine the anti knock characteristics of gasoline

Octane Number

- Draws fuel from tank to the primary fuel filter, this provides flow throughout the low portion of the fuel system.

Fuel Transfer Pump

- Acts as an balancer and provide momentum during dead stroke in a cycle

Flywheel

- Are steel tubes with sit at both end and bridge the motion from cam shaft to rocker arm

Push rod

- Passage of coolant from the engine block to the cylinder head.

Water jacket

- Are used to cool incoming air so that the volume of air available is inc.

After cooler

- The difference between the maximum and minimum volume.

Displacement Volume

- This are used to seal the gasses within the cylinder and to keep oil out

Piston Rings

- Process where heat is added in the Otto cycle

Constant Volume

- Process where heat is added in the Diesel cycle

Constant Pressure

- It is used to reduce the piston bearings and sliding surfaces in machine and does diminish the wear, heat and possibility of seizure of the parts.

Lubricant

- In a heat engine the ratio of brake power to the indicated power is called

Mechanical Efficiency

- The efficiency of Otto cycle depends upon the

Compression Ratio

- The maximum temperature of diesel cycle will occur.

At the end of Isobaric heating

- The ratio of the average load to the peak load over the designated period of time.

Load Factor

- The ratio of the sum of the individual maximum demands of the system to the overall maximum demands of the whole system.

Diversity Factor

- The sum of continuous ratings of all equipments and outlets on the customers circuits

Connected Load

- The ratio of the duration of the actual service of the machine or equipment to the total duration of a period of time considered is called.

Operational Factor

- The ratio of the break mean effective pressure to the indicated mean effective pressure

Mechanical Efficiency

- The ratio of the combined engine efficiency to the brake engine efficiency.

Generator Efficiency

- The ratio of the average load to that of the peak load of a plant is called:

Load Factor

- The ratio of the peak load to the connected load is called:

Capacity Factor

- The difference between the power plant installed capacity and the peak load is called.

Reserved Peak Load

- A series of processes during when the initial state point and the final state point are the same:

Cycle

- The compressibility factor of an ideal gas is equal to:

1.00

- The relation $PV=C$ represents a process or change of state which is known as

Isothermal Process

- In the relation $PV^n = C$, if the value of $n = 0$ the process is said to be:

Isobaric Compression

- In a carnot cycle heat rejection is the _____ process.

Isothermal Compression

- The system in which the mass flow rate out flow are not equal or vary with time and in which the mass within the system changes with time.

Unsteady Flow System

- A form of energy that is a sole function of temperature for perfect gasses and strong function of temp. and weak function of pressure for non perfect gasses, vapors and liquids.

Internal Energy

- It is that portion of the universe an atom a certain quantity of matter or a certain volume in space is that one wishes to study.

System

- It is the force of gravity per unit volume of a substance.

Specific Weight

- It is a kind of thermodynamic system whose mass does not cross its boundaries

Closed System

- Which laws states that acceleration of a particular body is directly proportional to the resultant force acting on it and inversely proportional to its mass.

Newtons 2nd Law of Motion

- A substance that receives transport and transfers energy

Working Substance

- A system where energy and mass cross its boundaries

Isolated System

- The Law of thermodynamics that deals with the law of conservation of energy which states that energy can neither be created or destroyed

First Law of Thermodynamics

- It states that in any mechanical mixture of gasses and vapors those that do not combined chemically the total pressure of the gaseous mixture such as moist air is equal to the sum of the partial pressure exerted by the individual gasses or vapors.

Daltons Law

- It is a substance existing in the gaseous phase of relatively near its saturation temp.

Vapor

- The point at which the saturated liquid and saturated vapor states are identical

Critical Point

- The point at which heat transfer stops

Thermal Equilibrium

- It is an instrument used for determining the specific gravity of a solution.

Hydrometer

- It refers to the temp. At which all molecules motion ceases according to the kinetic theory of heat.

0 K, R, -273,
All of the above

- The area under a curve on a pressure volume diagram represents

Non flow work

- A boiler steam pressure gage should have a range of at least _____ times the max allowable working pressure.

1.50

- It is a valve designed to allow a fluid to pass through in one direction only

Check Valve

- An ideal gas is compressed isothermally the enthalpy change is equal to

Zero

- The _____ is constant in an adiabatic throttling process

Enthalpy

- It is a the ideal cycle for gas turbine plant

Brayton Cycle

- It is the most thermal efficient cycle consisting of two isentropic process and two isothermal process.

Carnot cycle

- How can the average temperature during heat rejection process of a rankine cycle be decreased?

Reduce Turbine Exit Pressure

- What is commonly done to a vapor power cycle when the turbine has excessive moisture

Reheating

- A form of energy that is transferred between two system by virtue of temperature difference

HEAT

- A thermodynamic process with no heat transfer

Isentropic process

Adiabatic process

Throttling process

All of the above

- An energy interaction that is not caused by a temperature difference between a system and its surroundings

Work

- Changing of solid directly to vapor without passing through the liquid state is called.

Sublimation

- A rigid container is heated by the sun, there is no shaft work associated with the container. From the first law of thermodynamics, you determine the resulting work to be

Equal to zero

- The network output of a heat engine is always _____.

Less than the amount of heat input

- The sum of all energies of all the molecules in the system, energy that appears in several complex forms.

Internal energy

- Is a liquid whose temperature below the saturation temperature corresponding to its pressure.

Subcooled liquid

- A device that violates the 1st law of thermodynamics is called a

Perpetual motion machine of the 1st kind

- The difference of the superheated temperature and saturation temperature at that pressure is known as..

Degree of superheat

- The parsons reaction turbine has.

Identical fixed and moving blades

- The maximum efficiency of the de - laval turbine in terms of nozzle angles is

$\cos^2 \alpha$

- Curtis turbine is a

Pressure velocity compounded turbine

- The compounding of turbine is done in order to

Reduce speed rotor

- The Reheat factor is the ratio of

Commulative heat drop to the isentropic heat drop

- The function of a distributor in a coil ignition system of internal combustion engine is

To time to spark

- The knocking tendency in spark ignition engine maybe decreased by

Reducing the compression ratio

- The volumetric efficiency of a well designed engine

75% to 90 %

- The morse test is used to find the indicated power of a

Multi cylinder engine

- The volume of air delivered by the compressor is called

Compressor capacity

- The rotary compressors are used for delivering

Large quantities of air at low pressure

- In a centrifugal compressors, an increased in speed at a given pressure ratio causes

Increase in flow and decrease in efficiency

- Which of the following statement is correct as regard to centrifugal compressor

The static pressure of air in the impeller increases in order to provide centripetal force on the air

- The volumetric efficiency of compressor

Decreases with increase in compression ratio

- The sub-cooling is a process of cooling the refrigerant in vapor compression refrigeration system

Before throttling

- When food products or other hygroscopic material kept in different compartments are to be maintained at same temperature then the evaporator

None of these

- The refrigerant commonly used in vapor absorption system is.

Aqua ammonia

- A vapor absorption system

Gives quiet operation

- An electrolux refrigerator is called a

Three fluids absorption system

- For steam

The critical temperature is 374.15°C and critical pressure is 221.2 bar

- With the increase in pressure

The boiling point of water increase and enthalpy of vaporization decreases

- The specific volume of water when heated from 0°C

First decreases and then increases

- The fuel mostly used in boilers is

Non - caking bituminous coal

- Steam coal is a

Non - caking bituminous coal

- A process of heating crude oil to a high temperature under a very high pressure to increase the yield of lighter distillates is known as

Cracking

- Which of the following statement is incorrect

The solid fuel have higher efficiency than liquid fuel

- 1kg of carbon monoxide requires $4\frac{7}{7}$ kg of oxygen and produces

$7\frac{7}{7}$ kg CO₂

- 1kg of ethylene requires 2kg of carbon oxygen and produces $22\frac{7}{7}$ kg of carbon dioxide

$9\frac{7}{7}$ kg of water

- the mass of carbon per kg of flue gas is given by

$11\frac{1}{3}$ CO₂ + $3\frac{7}{7}$ CO

- A safety valve mainly used with locomotive and marine boiler is

Spring loaded safety valve

- A device used in boiler to control the flow of steam from the boiler to the main pipe and to shut off the steam completely when required is known as

Stop valve

- A device used to put off fire in the furnace of the boiler when the level of water in the boiler falls to an unsafe limit

Fusible plug

- In a boiler, various heat loss take place the biggest lost is due to

Dry flue gasses

- The draft in locomotive boilers is produce by a

Steam jet

- Which of the following statement is wrong

The natural draft reduces the fuel consumption

- The average value of diagram factor lies between

0.65 to 0.9

- The missing quality per stroke is equal to

Cylinder feed - indicated mass of steam

- The ratio of actual vacuum to the ideal vacuum in a condenser is

Vacuum efficiency

- The flow of steam is supersonic

In the divergent portion of the nozzle

- Comparing the mechanical vapor compression refrigeration system with absorption refrigeration the compressor of the former is the case in the latter by

An absorber, a generator, liquid pump and a pressure reduction valve

- The freon group of refrigerants are

Halocarbon refrigerants

- A refrigerant with the highest critical temperature is

Ammonia

- Which of the following is an azeotropes refrigerants

R- 502

- The color of the flame of halide torch incase leakage of freon refrigerant will change to

Bright green

- Which of the following refrigerant is highly toxic and flammable

Ammonia

- In large industrial and commercial reciprocating compression system, the refrigerant widely used is

Ammonia

- Environmental protection agencies advise against the used of chloroflourocarbon refrigerant since

These reacts with ozone layer

- Most aircooled condensers are designed to operate w/ a temperature difference of **14°C**

- The pressure in the capillary tube due to **fractional resistance offered by the tube wall acceleration of the refrigerant in the tube**
Ans. Both A and B

- Capillary tube is not used in large capacity refrigeration system because **Capacity control is not possible**

- The thermostatic expansion valve operates on the change in the

Degree of superheat at exit from the evaporator

- Most thermostatic expansion valve are set for superheat of

5°C

- The process of removing the moisture from the food product is called

Dehydration

- In air blast freezing method

A very low temp. air is circulated w/ a very high velocity

- In air blast freezing the velocity of air varies from

30 to 120 m/min

- The difference between the DB temp and WB

WB Depression

- The WB depression is zero when the relative humidity is equal to

1.0

- The minimum temp to which moist air can be cooled under ideal conditions in a spray washer is

WB Temperature of inlet air

- The bypass factor of a cooling decreases with

Decreases in sin spacing & increase in no. of rows

- In order to cool and dehumidify a steam moist air it must passed over a coil at the temp

Which is lower than the dew point temp of the incoming steam

- The index which correlates the combined effects of air temp relative humidity and air velocity on the human body is known as

Effective Temperature

- For rectangular duct the aspect ratio is equal to

Ration of longer and shorter sides

- The air guide vains are sometimes installed in axial flow fans in order to

Eliminate spiral flow of discharge air

- The axial flow fans are particularly suitable for handling

Large volume of air at relative low pressures

- The room air conditioner controls the

Temperature and humidity of air

- The milky white ice is obtained if

Air is present in it

- The dry ice is produced by

By solidifying liquid CO₂

- The process of heating and immediately cooling the milk for controlling the bacteria growth is known as

Pasturization

- Ice of portable water can crack if frozen at a temp lower than

-12°C